

Source Transformation Theorem

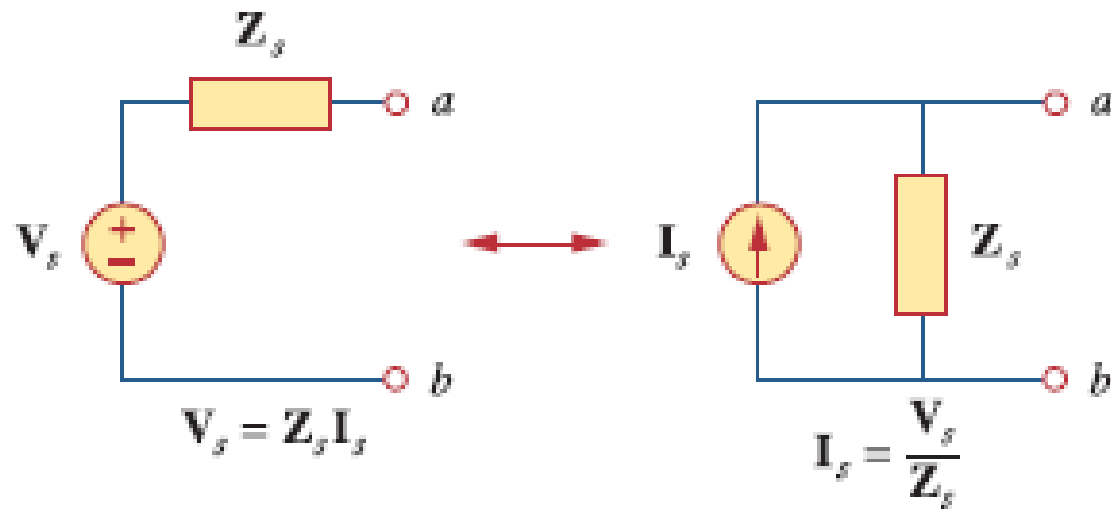
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Source Transformation Theorem

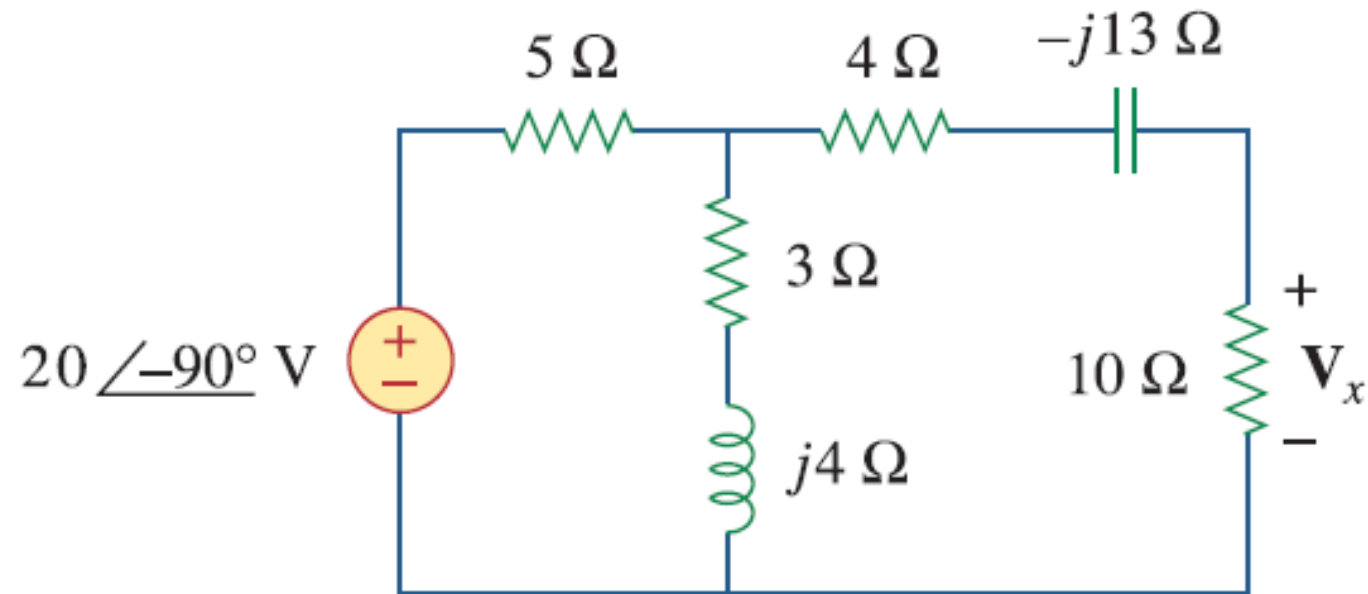
- Source transformation in the frequency domain involves transforming a voltage source in series with an impedance to a current source in parallel with an impedance, or vice versa.



$$V_s = Z_s I_s \quad \Leftrightarrow \quad I_s = \frac{V_s}{Z_s}$$

Numerical -1

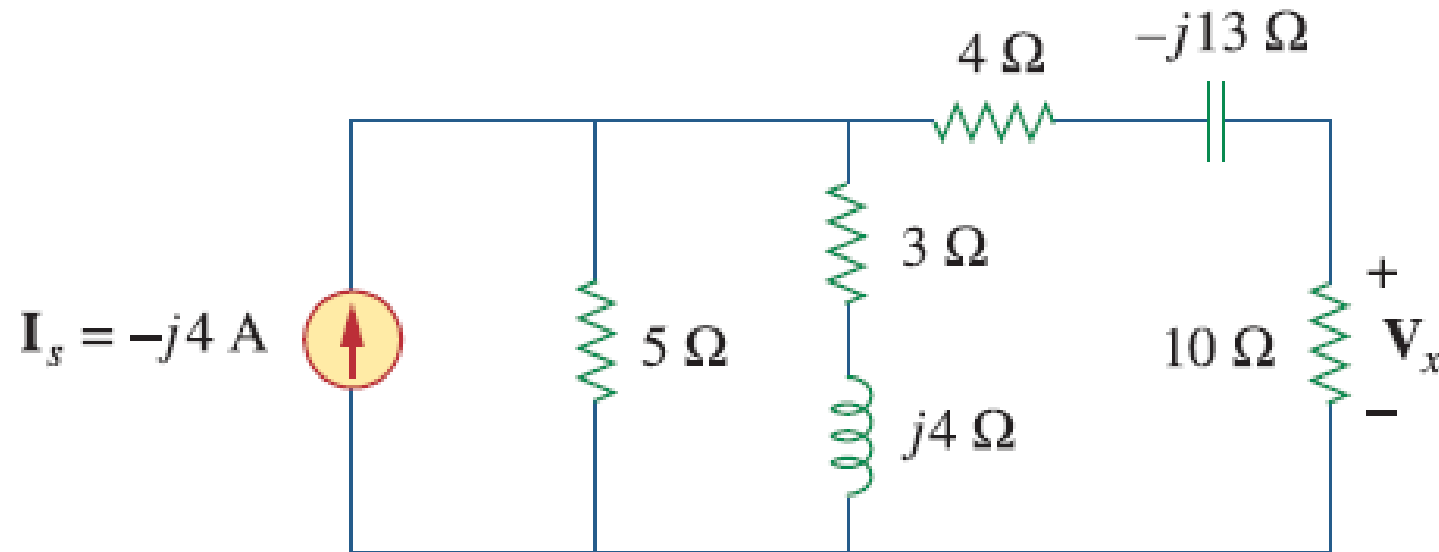
Calculate V_x in the circuit using the method of source transformation.



Solution:

We transform the voltage source to a current source

$$\mathbf{I}_s = \frac{20 \angle -90^\circ}{5} = 4 \angle -90^\circ = -j4 \text{ A}$$

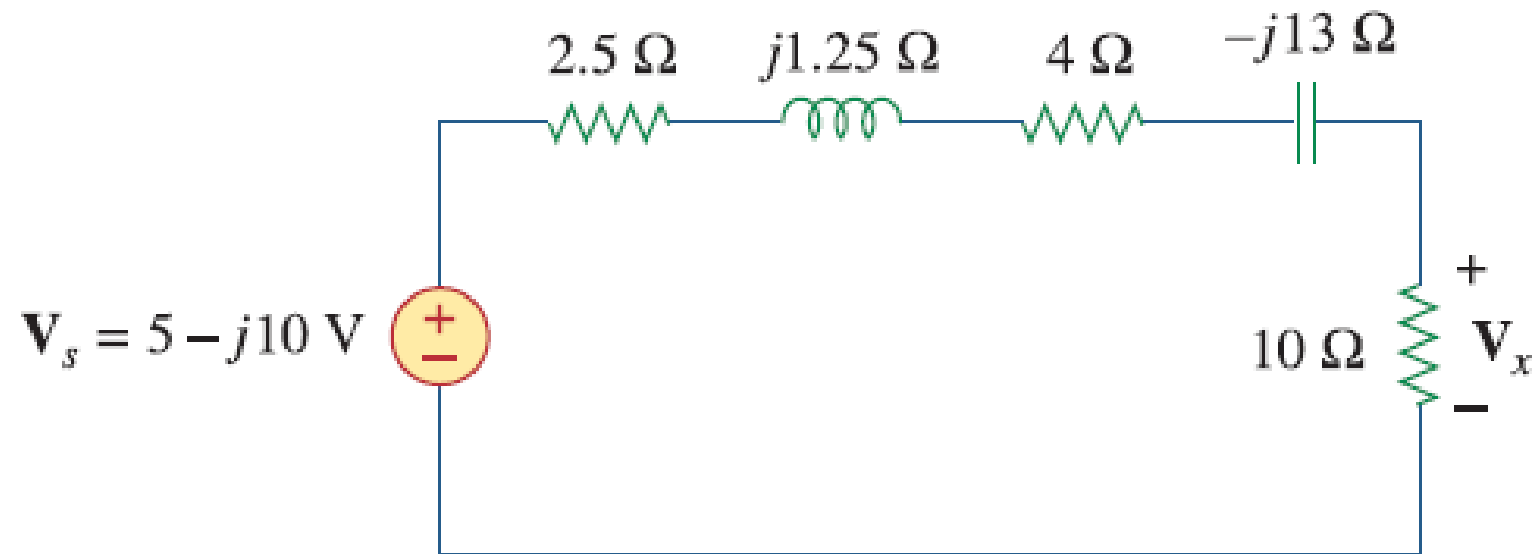


The parallel combination of $5\text{-}\Omega$ resistance and $(3 + j4)$ impedance gives

$$\mathbf{Z}_1 = \frac{5(3 + j4)}{8 + j4} = 2.5 + j1.25 \Omega$$

Converting the current source to a voltage source

$$\mathbf{V}_s = \mathbf{I}_s \mathbf{Z}_1 = -j4(2.5 + j1.25) = 5 - j10 \text{ V}$$

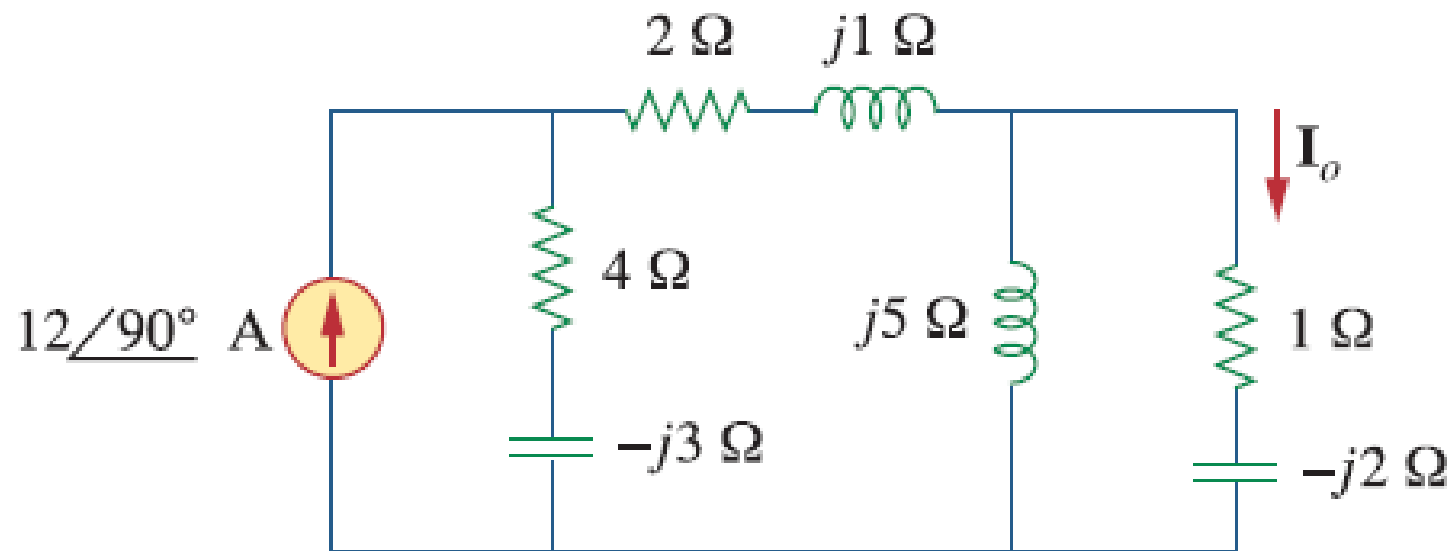


By voltage division,

$$\mathbf{V}_x = \frac{10}{10 + 2.5 + j1.25 + 4 - j13} (5 - j10) = 5.519 \angle -28^\circ \text{ V}$$

Numerical - 2

Find $\mathbf{I_o}$ in the circuit using the concept of source transformation.



Answer: $9.863\angle 99.46^\circ$ A.